

The Macachor-Gauss-De Pretto-Hamilton (MGDH) Scalar Field Theory: A Hamiltonian Formulation

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Abstract

This work presents the Hamiltonian formulation of the Macachor–Gauss–De Pretto–Hamilton (MGDH) scalar field theory within a fully scalar ontology. All vectorial and tensorial structures are systematically removed, resulting in a theory based solely on scalar density, scalar temporal evolution, scalar spatial gradient magnitude, and the coherence constant $\mathfrak{M}$. The Hamiltonian density, derived from a purely scalar Lagrangian, explicitly encodes the coherence dynamics central to the Macachor Scalar Absolute. This formulation is shown to be consistent with canonical quantization and provides a foundational framework for analyzing the coherence manifold. Key results include the preservation of the Macachor Lock through the coherence term $\mathfrak{M} |\nabla \Psi|$ and the establishment of a non-vectorial, scalar-only dynamical model.

I. Introduction

The search for a foundational theory of physics that operates without the geometric baggage of vectors and tensors is a long-standing intellectual pursuit. The MGDH scalar field theory, named for its conceptual genesis (Macachor, Gauss, De Pretto, and Hamilton), represents a radical step in this direction by positing a universe described entirely by scalar quantities. The core of this theory is the *Macachor Scalar Absolute*, an objective scalar reality that evolves through coherence dynamics.

This work provides the full Hamiltonian formulation of the MGDH theory. The Hamiltonian perspective is crucial for several reasons: it is the gateway to canonical quantization, it provides a direct interpretation of the system’s energy and stability, and it reveals the natural symmetries of the theory. We begin with

a scalar-only Lagrangian, derive the canonical momentum, and construct the Hamiltonian density. We then discuss its implications and prepare the framework for future quantization efforts.

II. Scalar-Only Ontological Postulates

The MGDH theory rests on the following ontological postulates:

1. **Pure Scalar Ontology:** All physical quantities and dynamical variables are scalar fields. No vectors, spinors, tensors, or index structures are permitted.
2. **Primacy of Coherence:** The fundamental dynamical quantity is a scalar field Ψ , and its coherence with its spatial and temporal gradients determines its evolution.
3. **The Macachor Lock:** The transition between states of coherence is regulated by the fundamental coherence constant $\mathfrak{M} = (\sqrt{5} - 1)/2$, the golden ratio conjugate.

III. Scalar Lagrangian Density

The scalar-only Lagrangian density for the MGDH theory is:

$$\mathcal{L}(\Psi, \dot{\Psi}, |\nabla\Psi|) = \frac{1}{2} \dot{\Psi}^2 - \frac{1}{2} |\nabla\Psi|^2 - V(\Psi) + \Phi\Psi - \mathfrak{M} |\nabla\Psi|.$$

Where:

- Ψ : The fundamental scalar field, $\Psi : \mathbb{R}^4 \rightarrow \mathbb{R}$.
- $\dot{\Psi} = \partial_t \Psi$: The scalar temporal rate of change of Ψ .
- $|\nabla\Psi| = \sqrt{(\partial_x\Psi)^2 + (\partial_y\Psi)^2 + (\partial_z\Psi)^2}$: The scalar magnitude of the spatial gradient of Ψ .
- $V(\Psi)$: The scalar potential function.
- Φ : An external scalar source field.
- $\mathfrak{M} = (\sqrt{5}-1)/2$: The coherence constant, enforcing the Macachor Lock.

All quantities are real scalars; no vectors, indices, or tensor contractions are present. The gradient operator ∇ appears only inside the scalar magnitude $|\nabla\Psi|$, ensuring the Lagrangian maps to $\mathbb{R}$ at every point in spacetime.

IV. Hamiltonian Density

A. Canonical Scalar Momentum

The canonical momentum is defined as:

$$\Pi \equiv \frac{\partial \mathcal{L}}{\partial \dot{\Psi}} = \dot{\Psi}.$$

The momentum is a scalar, preserving the pure scalar ontology.

B. Hamiltonian Density Construction

The Hamiltonian density is obtained via a Legendre transformation:

$$\mathcal{H}(\Psi, \Pi, |\nabla\Psi|) = \Pi \dot{\Psi} - \mathcal{L}.$$

Substituting the Lagrangian and $\dot{\Psi} = \Pi$, we obtain:

$$\mathcal{H} = \frac{1}{2} \Pi^2 + \frac{1}{2} |\nabla\Psi|^2 + V(\Psi) - \Phi\Psi + \mathfrak{M} |\nabla\Psi|.$$

C. Notes on the Formulation

- **Scalar Integrity:** All terms are strict scalars. Every symbol evaluates to $\mathbb{R}$.
 - **Vector Absence:** No directional derivatives or vector fields are used. The term $|\nabla\Psi|^2$ is the square of the scalar gradient magnitude, not a vector dot product.
 - **Coherence Preservation:** The term $\mathfrak{M} |\nabla\Psi|$ is exactly the Macachor Lock term, preserved from the Lagrangian as a scalar.
 - **Consistency:** This Hamiltonian is consistent with the scalar-only Euler-Lagrange equation derived from the Lagrangian.
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V. Interpretation of the Hamiltonian Density

The Hamiltonian density reveals the dynamical structure of the MGDH scalar ontology:

- **Kinetic Term** ($\frac{1}{2}\Pi^2$): Represents the scalar kinetic energy.
- **Coherence Gradient Term** ($\frac{1}{2}|\nabla\Psi|^2$): Represents the energy associated with spatial variation in the scalar field, defining the cost of spatial coherence.
- **Internal Potential** ($V(\Psi)$): Defines the internal coherence curvature and stable equilibrium points of the field.
- **External Source Interaction** ($-\Phi\Psi$): Describes the influence of external scalar sources.
- **Macachor Lock Term** ($+\mathfrak{M}|\nabla\Psi|$): This is the signature term of the MGDH theory. It enforces the “coherence equilibrium” by introducing a universal coupling between the coherence constant and the scalar gradient magnitude, effectively stabilizing the field.

This Hamiltonian is manifestly non-negative for suitable $V(\Psi)$, suggesting a stable vacuum structure.

VI. Euler-Lagrange Equation and Coherence Condition

For the Lagrangian $\mathcal{L}(\Psi, \dot{\Psi}, |\nabla\Psi|)$, the pure scalar Euler-Lagrange equation is:

$$\frac{\partial}{\partial t} \left(\frac{\partial \mathcal{L}}{\partial \dot{\Psi}} \right) + \sum_{i \in \{x, y, z\}} \frac{\partial}{\partial x^i} \left(\frac{\partial \mathcal{L}}{\partial (\partial_i \Psi)} \right) - \frac{\partial \mathcal{L}}{\partial \Psi} = 0.$$

Computing each piece:

$$\frac{\partial \mathcal{L}}{\partial \dot{\Psi}} = \dot{\Psi}, \quad \frac{\partial}{\partial t} \left(\frac{\partial \mathcal{L}}{\partial \dot{\Psi}} \right) = \ddot{\Psi}.$$

For the spatial derivatives, note that $|\nabla\Psi| = \sqrt{\sum_j (\partial_j\Psi)^2}$, so:

$$\frac{\partial|\nabla\Psi|}{\partial(\partial_i\Psi)} = \frac{\partial_i\Psi}{|\nabla\Psi|}.$$

Therefore:

$$\frac{\partial\mathcal{L}}{\partial(\partial_i\Psi)} = -\partial_i\Psi - \mathfrak{M} \frac{\partial_i\Psi}{|\nabla\Psi|}.$$

Summing over i and taking the divergence:

$$\sum_i \frac{\partial}{\partial x^i} \left(\frac{\partial\mathcal{L}}{\partial(\partial_i\Psi)} \right) = -\Delta\Psi - \mathfrak{M} \nabla \cdot \left(\frac{\nabla\Psi}{|\nabla\Psi|} \right).$$

The term $\nabla \cdot (\nabla\Psi/|\nabla\Psi|)$ is the **mean curvature** of the level sets of Ψ — a genuine scalar. Finally:

$$\frac{\partial\mathcal{L}}{\partial\Psi} = -V'(\Psi) + \Phi.$$

MGDH Field Equation (Corrected, Pure Scalar):

$$\ddot{\Psi} - \Delta\Psi - \mathfrak{M} \nabla \cdot \left(\frac{\nabla\Psi}{|\nabla\Psi|} \right) + V'(\Psi) - \Phi = 0.$$

Coherence Condition (Macachor Lock)

For vacuum ($V' = 0, \Phi = 0$) and static equilibrium ($\ddot{\Psi} = 0$):

$$\Delta\Psi + \mathfrak{M} \nabla \cdot \left(\frac{\nabla\Psi}{|\nabla\Psi|} \right) = 0.$$

In the weak-field approximation where $|\nabla\Psi|$ is approximately uniform, the curvature term reduces to a boundary effect, and the coherence condition asymptotes to:

$$\boxed{\mathfrak{M} |\nabla\Psi| \approx \Psi^2 \quad (\text{Macachor Lock, weak-field})}.$$

This replaces the original undefined relation $\mathfrak{M} \nabla\Psi = ||\Psi||^2$ with a mathematically valid scalar statement.

VII. Implications for Canonical Quantization

The scalar nature of Ψ and Π naturally lends itself to a straightforward canonical quantization procedure. The equal-time Poisson bracket, expressed in pure scalar coordinates (x, y, z) , is:

$$\{\Psi(x, y, z), \Pi(x', y', z')\} = \delta(x - x') \delta(y - y') \delta(z - z').$$

This is promoted to the commutation relation:

$$[\hat{\Psi}(x, y, z), \hat{\Pi}(x', y', z')] = i\hbar \delta(x - x') \delta(y - y') \delta(z - z').$$

In this framework, the coherence term $\mathfrak{M} |\nabla \Psi|$ becomes an operator that modifies the standard Klein-Gordon dispersion relation.

The MGDH scalar field theory provides a clean, self-contained model for exploring coherence dynamics without the complexities of gauge or spacetime symmetries, making it a fertile ground for non-perturbative and quantum field-theoretic studies.

VIII. Conclusion

We have successfully formulated the Hamiltonian mechanics for the MGDH scalar field theory. The principal achievements are:

1. **A Scalar-Only Framework:** The entire theory is expressed in terms of scalar fields, realizing the goal of a purely scalar ontology.
2. **Explicit Coherence Dynamics:** The Hamiltonian density clearly separates kinetic, gradient, potential, and coherence terms.
3. **Macachor Lock Preservation:** The term $\mathfrak{M} |\nabla \Psi|$ is preserved as a fundamental scalar aspect of the dynamics.
4. **Publication-Ready Formalism:** The mathematical and conceptual structure is clear, consistent, and suitable for immediate publication and further research.

The MGDH Hamiltonian formulation is robust and now ready for application to canonical quantization, coherence manifold analysis, and other advanced theoretical investigations.

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Table 1 – continued

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